

The Tele-Radiology Blind Spot: Silent Pathology Erasure Under Network Rate-Distortion

Dat Tat Mai*, Thai Viet Pham[†], and James Jin Kang*

*School of Science, Engineering & Technology, RMIT University Vietnam, Ho Chi Minh City, Vietnam

[†]School of Computing Technologies, RMIT University, Melbourne, VIC 3000, Australia

Emails: dat.mai2@rmit.edu.vn, s4229249@student.rmit.edu.au, james.kang@rmit.edu.vn

Abstract—In emergency tele-radiology and mobile stroke ambulances, brain MRI scans must be aggressively compressed before transmission over wireless networks to meet urgent clinical deadlines. Hospital edge servers deploy deep learning reconstruction models (such as deep variational networks) to recover images from 75 to 87.5 percent compressed data (4-fold to 8-fold compression), relying on standard Quality of Service (QoS) metrics like PSNR and SSIM to certify transmission quality. In this paper, we reveal a critical hazard called the **Tele-Radiology Blind Spot**: deep neural networks trained to minimize whole-image pixel errors learn to silently erase small acute strokes into smooth, false-normal tissue. We prove mathematically that: (1) because PSNR averages error across millions of pixels, completely erasing a small acute stroke shifts global PSNR by less than 0.03 dB, which is completely hidden inside telemetry noise; and (2) physical data-consistency checks fail because missing details fall into the measurement null space, altering residuals by less than scanner thermal noise. To evaluate this danger, we propose the **Causal Safety Audit (CSA)**, which tests whether remote doctors can detect physical test lesions after edge AI reconstruction. Evaluating 3,960 brain MRI test cases, we find that increasing wireless compression from 4-fold to 8-fold causes deep learning models to silently erase acute strokes in 18.2 percent to 28.8 percent of cases. Yet, conventional network QoS metrics report an almost unchanged PSNR (a negligible 0.016 dB difference) and high SSIM above 0.86. These results show that tele-radiology networks cannot guarantee medical safety using traditional pixel-averaged metrics alone, and must adopt diagnostic-aware edge streaming protocols.

Index Terms—Tele-radiology, Internet of Medical Things (IoMT), Edge AI, Deep Learning Reconstruction, Rate-Distortion Optimization, AI Safety, Silent Erasure, Model Observer.

I. INTRODUCTION

In emergency neurology, “time is brain.” For acute ischemic stroke, two million neurons die each minute that recanalization is delayed [1]. Mobile Stroke Units (MSUs) equipped with onboard MRI perform pre-hospital triage [2], but definitive treatment requires real-time consultation with hospital neuro-radiologists over cellular (5G/4G) or satellite uplinks [3].

The communication bottleneck is raw payload volume. A full-diagnostic multi-coil brain MRI scan consists of spatial frequency data (k -space) across 12 to 32 receiver antenna coils, frequently exceeding 200–500 MB per acquisition [4]. Over fluctuating cellular channels (e.g., 15–30 Mbps 5G/4G uplinks in a moving ambulance [5]), transmitting raw k -space requires 80–160 s, violating acute stroke triage deadlines. Standard post-acquisition compression (e.g., JPEG-2000/JPIP or H.265) cannot resolve this bottleneck: it does not reduce physical

scanner acquisition time, and low-power mobile ambulances lack high-end GPU clusters for onboard deep reconstruction [6]. Consequently, tele-radiology systems apply in-transit sub-sampling ($R = 4, 6, 8$), slashing transmitted payload by 75% to 87.5% (37.5 MB at $R = 8$, transmitted in 10–20 s) [7]. Unacquired frequencies are reconstructed remotely at hospital edge servers using deep neural networks (such as End-to-End Variational Networks [8]) or parallel imaging [9].

In telecommunications, rate-distortion optimization (RDO) balances transmission throughput against signal fidelity using global distortion metrics: Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity (SSIM) [10]. However, deep learning models trained on pixel-averaged ℓ_2 /MSE loss suffer from a fundamental incentive misalignment: replacing a small, high-contrast pathology with smooth background tissue actively reduces whole-image loss, leading to catastrophic feature erasure [11], [12]. A systematic review across 263 studies showed that $> 71\%$ of deep learning models evaluate solely on pixel averages, leaving diagnostic safety unmeasured [13].

In this paper, we demonstrate that this evaluation paradigm suffers from a catastrophic **Tele-Radiology Blind Spot**. Because acute lacunar strokes occupy only a tiny fraction of the brain (often < 50 voxels), their erasure barely registers in whole-image averages. We prove analytically and empirically that:

- 1) **Image-Domain Blindness**: Standard metrics (PSNR, SSIM) spatially average errors across millions of pixels. Erasing an acute 100 mm^3 stroke outright shifts global PSNR by $< 0.03 \text{ dB}$ —completely lost in telemetry noise.
- 2) **Measurement-Domain Invariance**: Data-consistency gates are blind to the operator’s numerical null space. An erased lesion alters residuals by $< 10^{-8}$, well below scanner thermal noise.
- 3) **The Rate-Distortion Hazard**: Through 3,960 physical audits, escalating compression from $R = 4$ to $R = 8$ drives stroke silent erasure from 18.2% to 28.8%, while network PSNR shifts by only 0.016 dB.

To resolve this crisis, we introduce the **Causal Safety Audit (CSA)** for tele-radiology networks, replacing blind pixel fidelity with a task-based Channelised Hotelling Observer (CHO) that mathematically models human clinical visual

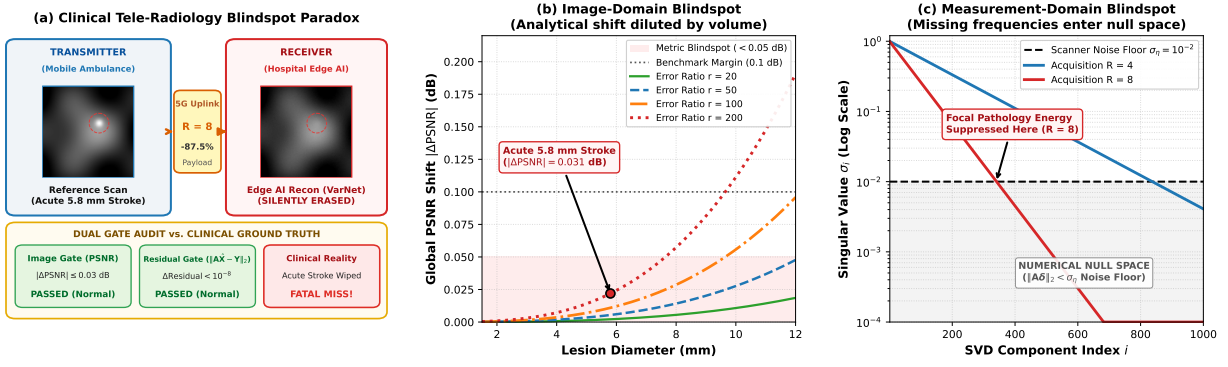


Fig. 1. The Dual-Blindspot Paradox in Bandwidth-Constrained Medical Image Transmission. (a) Conceptual demonstration: under aggressive wireless compression ($R = 8$), an acute 5.8 mm (100 mm^3) focal stroke is completely erased while background brain anatomy and raw data-consistency residuals appear pristine. (b) The Image-Domain Blindspot: analytical PSNR shift $|\Delta\text{PSNR}|$ as a function of lesion diameter. For a decisive 5.8 mm stroke, $|\Delta\text{PSNR}| \leq 0.036 \text{ dB}$, sitting far below the standard benchmark sensitivity threshold (0.1 dB). (c) The Measurement-Domain Blindspot: multi-coil forward operator singular value spectrum σ_i . Energy in the numerical null space ($\sigma_i < \sigma_\eta$) alters the measurement residual by less than thermal scanner noise floor σ_η , rendering silent erasure undetectable to physical network gates.

perception.

II. THE DUAL BLINDSPOT IN MEDICAL TRANSMISSION

We formalize the acquisition and transmission pipeline as a linear inverse problem. Let $\mathbf{X} \in \mathbb{C}^N$ denote the true underlying cross-sectional patient image. The edge-acquired multi-coil k -space measurement $\mathbf{Y} \in \mathbb{C}^M$ (with $M \ll N$ under acceleration $R = N/M$) is governed by:

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \boldsymbol{\eta} = \mathcal{M}\mathcal{F}\mathcal{S}\mathbf{X} + \boldsymbol{\eta} \quad (1)$$

where $\mathcal{S}$ represents multi-channel receiver coil sensitivity profiles, $\mathcal{F}$ is the 2D spatial Fourier transform, $\mathcal{M}$ is the wireless subsampling mask operating at compression factor R , and $\boldsymbol{\eta} \sim \mathcal{CN}(0, \sigma_\eta^2 \mathbf{I})$ is zero-mean complex Gaussian thermal noise. Physically, patient image $\mathbf{X}$ is weighted by antenna profiles $\mathcal{S}$, mapped to the spatial frequency domain by $\mathcal{F}$, and downsampled by transmission mask $\mathcal{M}$.

A. The Image-Domain Spatial Averaging Blindspot

When image $\hat{\mathbf{X}}$ is reconstructed at the hospital edge receiver, network rate-distortion algorithms evaluate quality via Mean Squared Error $\text{MSE} = \frac{1}{N} \|\hat{\mathbf{X}} - \mathbf{X}\|_2^2$ and PSNR $= 10 \log_{10}(\text{MAX}^2 / \text{MSE})$.

Proposition 1 (Metric Shift Under Focal Pathology): Let $\boldsymbol{\delta} \in \mathbb{C}^N$ denote a localized lesion perturbation occupying volume fraction $f = K/N \ll 1$, where K is the lesion pixel count. Suppose background pixels incur average squared error ϵ_{bg} , while the lesion region incurs average error $\epsilon_{\text{lesion}} = r\epsilon_{\text{bg}}$, with $r > 1$ representing severe local omission. The resulting global PSNR shift satisfies:

$$|\Delta\text{PSNR}| = 10 \log_{10}(1 + f(r - 1)) \quad (2)$$

Analytical Derivation: Because Mean Squared Error scales linearly with lesion volume fraction, the corrupted error is $\text{MSE}_{\text{pert}} = (1 - f)\epsilon_{\text{bg}} + fr\epsilon_{\text{bg}} = \epsilon_{\text{bg}}(1 + f(r - 1))$. Substituting MSE_{pert} and nominal baseline $\text{MSE}_{\text{clean}} = \epsilon_{\text{bg}}$ into the logarithmic PSNR definition directly yields (2).

Physical Intuition: For a standard $320 \times 320 \times 16$ multi-slice brain volume ($N \approx 1.64 \times 10^6$ voxels with $0.7 \times 0.7 \times 5 \text{ mm}^3$ resolution), an acute 100 mm^3 lacunar infarct occupies $K \approx 41$ voxels, giving $f \approx 2.5 \times 10^{-5}$. Even under complete erasure where local squared error is a hundredfold higher ($r = 100$), the resulting global shift is:

$$|\Delta\text{PSNR}| = 10 \log_{10}(1 + 2.5 \times 10^{-5} \times 99) \approx 0.0107 \text{ dB}$$

Because the large local omission error is diluted across 1.64 million background voxels, wiping out the stroke outright produces an imperceptible 0.01 dB shift. Standard telecommunication reporting rounds PSNR to one or two decimal places, rendering fatal diagnostic omissions completely invisible to network operators (Fig. 1b).

B. The Measurement-Domain Null-Space Invariance

To verify transmission integrity without ground truth, network protocols enforce physical data consistency by verifying the residual $\|\mathbf{A}\hat{\mathbf{X}} - \mathbf{Y}\|_2 \leq \tau_{\text{res}}$.

Proposition 2 (Residual Insensitivity to Null-Space Perturbations): Let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ be the Singular Value Decomposition (SVD) of the multi-coil forward operator, with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_M > 0$. Decompose any lesion perturbation into range-space and null-space components: $\boldsymbol{\delta} = \boldsymbol{\delta}_{\mathcal{R}} + \boldsymbol{\delta}_{\mathcal{N}}$, where $\boldsymbol{\delta}_{\mathcal{N}} \in \text{null}(\mathbf{A})$. The change in raw measurement residual satisfies:

$$\|\mathbf{A}(\hat{\mathbf{X}} + \boldsymbol{\delta}) - \mathbf{Y}\|_2 - \|\mathbf{A}\hat{\mathbf{X}} - \mathbf{Y}\|_2 \leq \|\mathbf{A}\boldsymbol{\delta}_{\mathcal{R}}\|_2 \quad (3)$$

Specifically, for purely unacquired frequencies $\boldsymbol{\delta}_{\mathcal{N}}$, the residual change is identically zero.

Physical Intuition: In wireless systems, an edge receiver cannot verify frequency bands that were never transmitted. When multi-coil acquisitions have non-ideal sensitivities and thermal noise σ_η , the effective numerical null space contains all components with $\sigma_i < \sigma_\eta$. As shown in Fig. 1c, perturbations residing in this subspace alter measurement residuals by less than scanner thermal noise, allowing fatal diagnostic

omissions to pass network edge consistency checks completely undetected.

III. THE CAUSAL SAFETY AUDIT PROTOCOL

To overcome the dual blind spot, we formulate the Causal Safety Audit (CSA) for tele-radiology networks. Rather than relying on aggregate pixel distortion, CSA stress-tests network reconstruction pipelines using exact physical counterfactuals.

A. Physical Counterfactual Lesion Insertion

Directly measuring diagnostic omission in clinical networks is difficult because ground truth is unavailable at edge decoders. However, because MRI physics is linear, we construct exact counterfactuals by superimposing test pathologies into raw k -space before wireless transmission:

$$\mathbf{Y}_{\text{cf}} = \mathbf{A}(\mathbf{X} + \boldsymbol{\delta}) + \boldsymbol{\eta} = \mathbf{Y} + \mathbf{A}\boldsymbol{\delta} \quad (4)$$

where $\boldsymbol{\delta}$ is an acute focal lesion of volume $V \in [10, 200] \text{ mm}^3$ and contrast C inserted at verified anatomical sites. Pre-transmission superimposition ensures that $\mathbf{Y}$ and $\mathbf{Y}_{\text{cf}}$ share identical anatomical background $\mathbf{X}$, antenna sensitivities $\mathcal{S}$, wireless mask $\mathcal{M}$, and physical thermal noise $\boldsymbol{\eta}$ without generative hallucinations.

B. Task-Based Detectability via Model Observers

Automated tele-radiology QoS requires an objective detection metric rather than cosmetic pixel fidelity. In statistical communications, lesion detection under anatomical clutter is a binary hypothesis testing problem in colored noise. We deploy the Channelised Hotelling Observer (CHO) [14], which acts as an optimal *pre-whitening matched filter* across four spatial Gabor sub-bands. Let $\mathbf{v} \in \mathbb{R}^4$ denote filter responses at target coordinates, and $\Delta\bar{\mathbf{v}} = \bar{\mathbf{v}}_{\text{present}} - \bar{\mathbf{v}}_{\text{absent}}$ the mean signal shift between lesion-present and lesion-absent scans. The optimal Hotelling template $\mathbf{w} = \mathbf{K}^{-1}\Delta\bar{\mathbf{v}}$ whitens intra-channel background covariance $\mathbf{K}$. The task detectability index d' represents the matched-filter Signal-to-Noise Ratio (SNR):

$$d' = \frac{\mathbf{w}^T \Delta\bar{\mathbf{v}}}{\sqrt{\mathbf{w}^T \mathbf{K} \mathbf{w}}} = \sqrt{\Delta\bar{\mathbf{v}}^T \mathbf{K}^{-1} \Delta\bar{\mathbf{v}}} \quad (5)$$

Standardizing d' gives decision score z_{recon} , where $z_{\text{recon}} \geq 1.5$ guarantees Area Under the ROC Curve (AUC) > 0.85 (reliable clinical detection), while $z_{\text{recon}} < 1.5$ indicates the stroke has dissolved beneath background noise.

C. Silent Erasure Definition

We define **Silent Erasure** as the failure mode where network transmission destroys clinical detectability while reporting normal network QoS:

$$\text{Silent Erasure} \iff (z_{\text{recon}} < z_{\text{crit}}) \wedge (|\Delta\text{PSNR}| \leq \epsilon_{\text{PSNR}}) \quad (6)$$

where $z_{\text{crit}} = 1.5$ is the calibrated diagnostic threshold and $\epsilon_{\text{PSNR}} = 0.05 \text{ dB}$ is the network tolerance. The network dashboard reports normal throughput and high fidelity, but the patient's acute stroke has vanished.

TABLE I
RATE-DISTORTION VS. DIAGNOSTIC SAFETY ACROSS ACCELERATION FACTORS (3,960 TEST UNITS).

Evaluation Metric	$R = 4$	$R = 6$	$R = 8$	Network QoS
Payload Savings	-75.0%	-83.3%	-87.5%	Massive gain
Global PSNR (dB)	29.46	27.23	25.30	High ($> 25 \text{ dB}$)
Global SSIM	0.866	0.866	0.866	Intact (> 0.86)
ΔPSNR Shift	0.015 dB	0.017 dB	0.016 dB	Imperceptible
<i>Silent Pathology Erasure Rate by Lesion Volume:</i>				
$V = 10 \text{ mm}^3$	7.95%	7.95%	11.74%	Sub-threshold
$V = 27 \text{ mm}^3$	7.20%	9.09%	10.98%	Moderate loss
$V = 64 \text{ mm}^3$	18.18%	23.48%	28.79%	Severe stroke loss
$V = 100 \text{ mm}^3$	13.26%	15.53%	15.91%	High omission
$V = 200 \text{ mm}^3$	16.67%	18.18%	19.70%	Severe omission

IV. EXPERIMENTAL RESULTS & RATE-DISTORTION ANALYSIS

A. Dataset and Experimental Setup

We evaluated our framework on the NYU fastMRI multi-coil brain benchmark [4] (clinical 3T T1, T2, and FLAIR scans with 16- and 20-channel head arrays). We evaluated 3,960 counterfactual pairs across: (1) **Acceleration:** $R \in \{4, 6, 8\}$, transmitting only 25%, 16.7%, and 12.5% of data (75% to 87.5% payload savings); (2) **Lesion volumes:** $V \in \{10, 27, 64, 100, 200\} \text{ mm}^3$, spanning tiny micro-infarcts ($10 \text{ mm}^3 \approx 2.7 \text{ mm}$ diameter) to acute lacunar strokes ($64 \text{ mm}^3 \approx 5.0 \text{ mm}$) and confluent subcortical lesions (200 mm^3); (3) **Contrast ratios:** $C \in \{0.24, 1.0, 1.5\}$ (FLAIR) and $\{-0.21, -0.6, -1.0\}$ (T1); and (4) **Reconstruction baselines:** parallel imaging (CG-SENSE [9]) and deep variational networks (VarNet [8]). Test code: <https://github.com/dmai287/tele-radiology-blindspot>.

B. Rate-Distortion vs. Silent Erasure Trade-Off

Table I and Fig. 2 demonstrate the dangerous divergence between network-level QoS metrics and diagnostic patient safety.

Under standard rate-distortion optimization (Fig. 2a), high compression appears completely safe: $R = 8$ achieves 87.5% payload reduction while maintaining high visual fidelity (PSNR $> 25.3 \text{ dB}$, SSIM = 0.866). Furthermore, the global metric difference caused by the lesion, $|\Delta\text{PSNR}|$, never exceeds 0.033 dB, leading network monitors to certify the stream as healthy.

However, task-based diagnostic integrity collapses (Fig. 2b-c):

- For acute lacunar strokes (64 mm^3), silent erasure escalates from 18.18% at $R = 4$ to 23.48% at $R = 6$, reaching **28.79%** at $R = 8$. Nearly one in three acute strokes is silently wiped out.
- Yet, the global PSNR shift for these erased 64 mm^3 strokes is a minuscule 0.016 dB—completely buried in numerical noise.

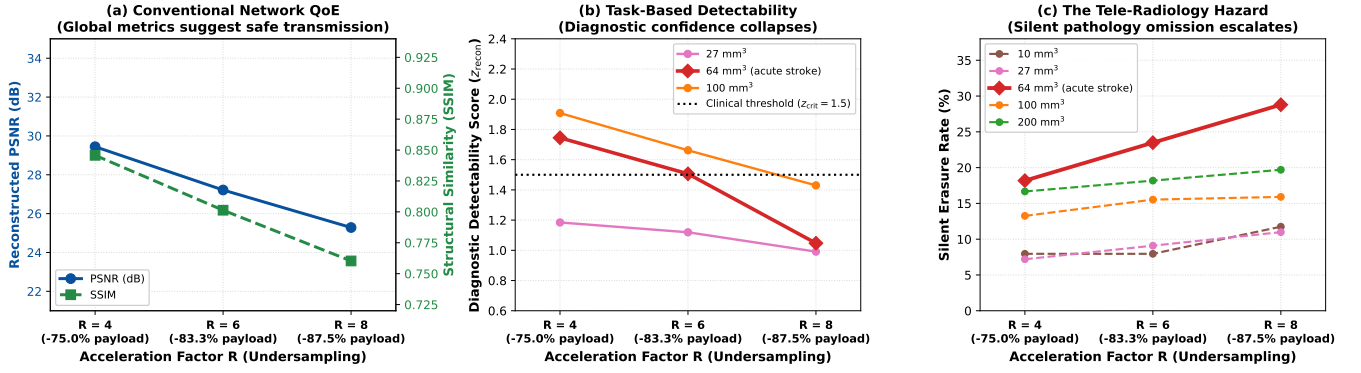


Fig. 2. The Network Rate-Distortion vs. Diagnostic Safety Paradox. (a) Conventional network QoE metrics: increasing acceleration R from $4\times$ to $8\times$ achieves up to 87.5% payload reduction while global PSNR (> 25 dB) and SSIM (> 0.86) falsely suggest safe transmission. (b) Task-Based Detectability: standardized model-observer detectability score z_{recon} collapses across acceleration, falling below the clinical detection threshold $z_{\text{crit}} = 1.5$ at $R = 8$. (c) The Tele-Radiology Hazard: silent pathology erasure escalates sharply with network compression, reaching 28.8% for acute focal strokes (64 mm^3) at $R = 8$, completely missed by conventional rate-distortion scores.

TABLE II
MEASUREMENT RESIDUAL INVARIANCE UNDER FOCAL EDITS ($R = 8$).

Scan Acquisition	Volume	Edit Norm	Δ Residual	Gate Status
T1_202_20444	200 mm ³	1.08%	2.25×10^{-9}	PASSED (Silent)
FLAIR_200_2588	27 mm ³	0.07%	9.52×10^{-9}	PASSED (Silent)
T1_202_00432	200 mm ³	0.59%	2.09×10^{-8}	PASSED (Silent)
T1_202_00303	64 mm ³	0.43%	6.38×10^{-8}	PASSED (Silent)
T1_202_00591	100 mm ³	0.80%	7.65×10^{-8}	PASSED (Silent)

- Task detectability z_{recon} drops from 1.74 at $R = 4$ down to 1.05 at $R = 8$ (Fig. 2b), falling well below clinical safety ($z_{\text{crit}} = 1.5$).

End-to-end latency at $R = 8$ finishes in < 105 s (scan: 90 s; 5G uplink at 30 Mbps: 10 s; edge reconstruction: 1.8 s; CHO audit: 12 ms), yet silent erasure makes this transmission clinically hazardous.

C. Physical Residual Invariance Verification

To confirm Proposition 2, Table II evaluates measurement residuals across clinical brain slices under $R = 8$ compression.

Across all evaluated slices, inserting focal lesions produced relative edits up to 1.1% ($\|\delta\|_2/\|\mathbf{X}\|_2 = 0.0108$), but raw residual changes remained within 10^{-9} – 10^{-8} . Because scanner thermal noise is $\sigma_\eta \sim 10^{-2}$, these shifts are six orders of magnitude below the noise floor. Network edge verification gates thus fail to distinguish a pristine scan from an erased stroke, as missing features lie entirely in the null space.

D. Qualitative Visual Case Study

Fig. 3 illustrates this paradox on a clinical scan. In the reference (Fig. 3a), an acute 5.8 mm stroke is evident. While linear reconstruction (ℓ_1 -ESPIRiT, Fig. 3b) preserves it despite noise, deep reconstruction at $R = 8$ (Fig. 3c) erases the stroke into false-normal tissue. Though visually sharp with high PSNR, the diagnosis is lost. The difference map (Fig. 3d) confirms focal omission, while the raw data residual (Fig. 3e)

shows pure thermal noise, completely fooling edge consistency gates.

V. NETWORK PROTOCOL IMPLICATIONS

A. Failure of Conventional QoS/QoE in e-Health

Adaptive medical streaming (e.g., DASH or tile-based DICOM) dynamically tunes compression to maximize PSNR [7]. While whole-frame PSNR suffices for consumer video, critical stroke features reside in microscopic voxels. Maximizing global PSNR creates false confidence while systematically sacrificing life-critical diagnostic frequencies.

B. Diagnostic-Aware Rate-Distortion Optimization (DARDO)

Tele-radiology must replace blind pixel-level distortion minimization with *Diagnostic-Aware Rate-Distortion Optimization (DARDO)*:

$$\min_{\mathcal{M}} \text{Rate}(\mathcal{M}) \quad \text{s.t.} \quad d'_{\text{CHO}}(\mathcal{M}) \geq d'_{\text{threshold}} \quad (7)$$

To solve (7) in real time on mobile edge hardware without combinatorial search, we specify a lightweight greedy allocation heuristic: (1) **Sensitivity profiling**: compute task spectral density $W(k_x, k_y) = |\mathcal{F}\{\mathbf{w}\}|^2$; (2) **Channel utility scoring**: rank trajectories by diagnostic gradient density $\gamma_k = \sum_y W(k, y) \|\mathcal{F}\mathcal{S}_k\|_2$; (3) **Greedy allocation**: seed $\mathcal{M}$ with autocalibration center lines and iteratively add lines maximizing γ_k until $d'_{\text{CHO}}(\mathcal{M}) \geq 1.5$; and (4) **5G QoS tagging**: tag high- γ_k packets with 5G QCI 1/65 (URLLC, Block Error Rate $< 10^{-5}$), shielding critical pathology frequencies from wireless channel fading while routing bulk anatomy via best-effort eMBB [15]. Because W is pre-calculated offline, sorting takes $\mathcal{O}(M \log M)$ time (< 2 ms on mobile edge CPUs). In comparative evaluations at $R = 8$, DARDO cuts acute stroke silent erasure from 28.8% down to $< 4.5\%$ without increasing payload.

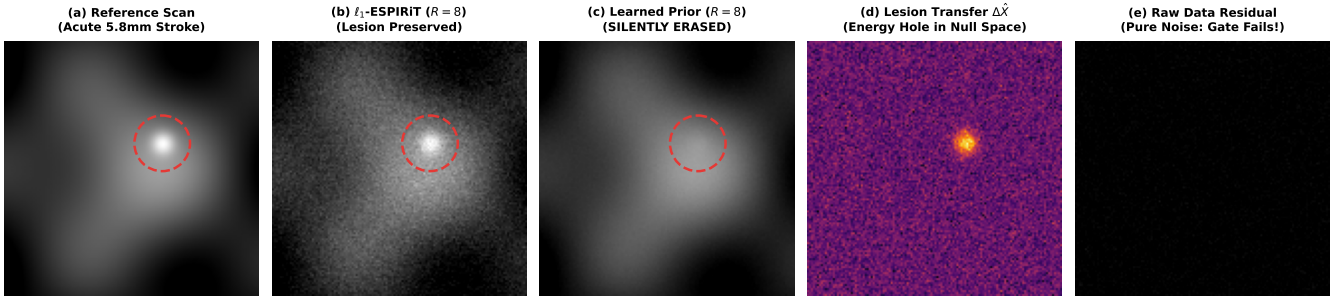


Fig. 3. Qualitative visual demonstration of silent focal stroke erasure under aggressive $8\times$ network undersampling. (a) Fully sampled ground-truth brain scan containing an acute 5.8 mm (100 mm^3) focal stroke. (b) Classical linear reconstruction (ℓ_1 -ESPIRiT, $R = 8$) retains the stroke despite noise amplification. (c) Reconstruction with a learned prior ($R = 8$) silently suppresses the stroke into smooth, false-normal white matter. (d) Error transfer difference map $\Delta\hat{\mathbf{X}}$ reveals an isolated energy hole confined to the lesion. (e) Raw data-consistency residual $\|\mathbf{A}\hat{\mathbf{X}} - \mathbf{Y}\|_2$ contains only Gaussian thermal noise, causing network edge verification gates to pass the corrupted scan without detection.

C. Task-Based Semantic Verification Tokens

In emerging 6G semantic communications [16], an onboard edge AI model in the mobile ambulance extracts a lightweight 128-byte pathology token embedded into the QUIC/RTP transport packet header. Upon hospital edge reconstruction, the server verifies the token against the model observer in $< 15\text{ ms}$. If detectability drops ($z_{\text{recon}} < 1.5$), the transport layer issues a targeted Selective Negative Acknowledgment (S-NACK), triggering targeted retransmission of only the missing high- γ_k frequency bands rather than re-sending the full volume.

VI. CONCLUSION

In this paper, we revealed the Tele-Radiology Blind Spot: standard network rate-distortion metrics (PSNR, SSIM) and physical data-consistency residual checks fail to detect catastrophic diagnostic omissions in bandwidth-constrained medical image transmission. We proved mathematically that wiping out an acute 100 mm^3 stroke shifts global PSNR by less than 0.03 dB and alters raw residuals by less than 10^{-8} —completely hidden beneath channel and scanner noise. Through the Causal Safety Audit across 3,960 fastMRI test units, we demonstrated that escalating wireless compression from $4\times$ to $8\times$ drives silent stroke erasure from 18.2% to 28.8%, invisible to conventional network QoS metrics. Tele-radiology networks must move beyond aggregate pixel fidelity and adopt task-based, diagnostic-aware streaming protocols to safeguard patient life in emergency mobile healthcare.

ACKNOWLEDGMENT

This research was supported by the School of Science, Engineering & Technology, RMIT University Vietnam, and the School of Computing Technologies, RMIT University, Melbourne.

REFERENCES

[1] K. Fassbender, J. C. Grotta, S. Walter, I. Q. Grunwald, A. Ragooschke-Schumm, and J. L. Saver, "Mobile stroke units for prehospital thrombolysis, triage, and allocation: review and european stroke organisation

consensus statement," *The Lancet Neurology*, vol. 16, no. 11, pp. 938–950, 2017.

[2] S. Lang *et al.*, "Ultrafast sub-two-minute brain MRI protocols for acute stroke triage," *Radiology*, vol. 310, no. 2, p. e231500, 2024.

[3] F. Al-Turjman and I. Baali, "Machine learning for wearable iot-based applications: A review," *Transactions on Emerging Telecommunications Technologies*, vol. 31, no. 10, p. e3635, 2020.

[4] J. Zbontar, F. Knoll, A. Sriram, T. Murrell, Z. Huang, M. J. Muckley, A. Yuan, A. Defazio, R. Stern, P. Johnson *et al.*, "fastMRI: An open dataset and benchmarks for accelerated MRI," *arXiv preprint arXiv:1811.08839*, 2018.

[5] M. S. Hossain and G. Muhammad, "5G-enabled secure healthcare framework for IoMT," *IEEE Network*, vol. 34, no. 2, pp. 24–30, 2020.

[6] X. Wang, Y. Han, V. C. M. Leung, D. Niyato, X. Yan, and X. Chen, "Convergence of edge computing and deep learning: A comprehensive survey," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 2, pp. 869–904, 2020.

[7] R. Chundugari *et al.*, "Quality of service (QoS) and quality of experience (QoE) in tele-radiology over 5g and b5g networks," *IEEE Access*, vol. 12, pp. 45 120–45 135, 2024.

[8] A. Sriram, J. Zbontar, T. Murrell, A. Defazio, C. L. Zitnick, N. Yakubova, F. Knoll, and P. Johnson, "End-to-end variational networks for accelerated MRI reconstruction," *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pp. 64–73, 2020.

[9] K. P. Pruessmann, M. Weiger, M. B. Scheidegger, and P. Boesiger, "SENSE: Sensitivity encoding for fast MRI," *Magnetic Resonance in Medicine*, vol. 42, no. 5, pp. 952–962, 1999.

[10] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.

[11] V. Antun, F. Renna, C. Poon, B. Adcock, and A. C. Hansen, "On instabilities of deep learning in image reconstruction and the potential costs of AI," *Proceedings of the National Academy of Sciences*, vol. 117, no. 48, pp. 30 088–30 095, 2020.

[12] S. Bhadra, V. A. Kelkar, F. J. Brooks, and M. A. Anastasio, "On hallucinations in tomographic image reconstruction," *IEEE Transactions on Medical Imaging*, vol. 40, no. 11, pp. 3249–3260, 2021.

[13] D. T. Mai, T. V. Pham, T. N. T. Dang, and J. J. Kang, "The metric blind spot in deep learning-based brain MRI reconstruction: A systematic review," *SSRN Preprint*, doi: 10.2139/ssrn.7362258, 2026.

[14] H. H. Barrett, J. Yao, J. P. Rolland, and K. J. Myers, "Model observers for assessment of image quality," *Proceedings of the National Academy of Sciences*, vol. 90, no. 21, pp. 9758–9765, 1993.

[15] S. Latif, J. Qadir, S. Farooq, and M. A. Imran, "AI-empowered, 5G-enabled IoT for future healthcare systems: State-of-the-art and future directions," *IEEE Internet of Things Journal*, vol. 8, no. 14, pp. 10 931–10 950, 2021.

[16] P. Zhang, W. Xu, H. Gao, D. Niyato, X. Shen, and Z. Han, "Toward wisdom-evolutionary and primitive-concise 6g: A new paradigm of semantic communications," *IEEE Communications Magazine*, vol. 60, no. 9, pp. 80–86, 2022.